Taking Prices as Given: Firms and Consumers in Competitive Markets

ECON 201: Principles of Microeconomics
Lecture 16

Div Bhagia

Last Class: Supply and Demand

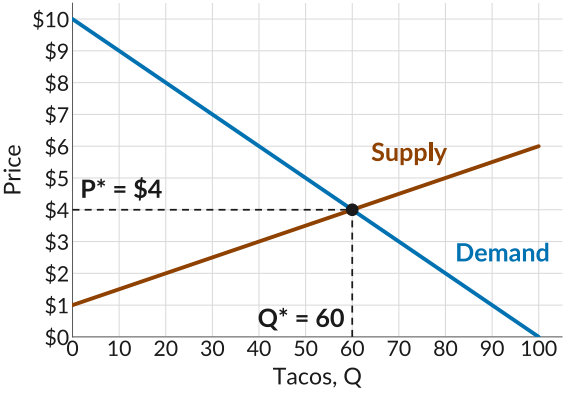
- In **competitive markets**, many sellers offer the same good, so they are **price-takers**: they sell at the going market price.
- **Demand**: how many units buyers want at each price. **Supply**: how many units sellers want to sell at each price.
- At the **market-clearing** or **equilibrium** price, the quantity demanded equals the quantity supplied, and nobody has an incentive to change.
- At any other price there is excess supply or excess demand, and sellers have an incentive to lower or raise their price.

Today: what it means to take the price as given, and how a firm decides how much to produce when it cannot choose its price.

Finding the Equilibrium With Algebra

Last class's taco curves are straight lines, so we can find where they cross with algebra.

- Demand: $P = 10 - Q/10$. Supply: $P = 1 + Q/20$.
- At the equilibrium both give the same price: $10 - Q/10 = 1 + Q/20$.
- So $9 = Q/10 + Q/20 = 3Q/20$, and $Q^* = 60$.
- Then $P^* = 10 - 60/10 = 4$. Supply gives $1 + 60/20 = 4$ too.



Price-Takers

A **price-taker** cannot gain by trading at any price other than the going market price. They cannot move the price, but they can buy or sell as much as they like at it.

- At $4, a stand asking $5 sells nothing, because shoppers go to the stands asking $4. A stand asking $3 gives money away, since shoppers will pay $4.
- A shopper offering $3 finds no seller, and one offering $5 overpays.
- Buyers are price-takers when there are many other buyers. Sellers are price-takers when many sellers offer an identical good. On both sides, competition removes bargaining power.

Competitive Equilibrium

- A market is in **competitive equilibrium** when the quantity supplied equals the quantity demanded at the going price, and every buyer and seller is a price-taker.
- It is a **Nash equilibrium**: given that everyone else trades at $4, no one can do better than trading at $4 too.
- The model needs two things. Many buyers and sellers, so that what any one of them trades is small next to the whole market. And identical goods, so that buyers can switch from one seller to another.

Price Setters and Price-Takers

Market	Sellers	Buyers
A differentiated product, e.g. Cheerios	**One seller, a price setter**. No one else sells Cheerios, so it can raise its price and still keep buyers.	**Many buyers, price-takers**. A buyer who offered less would be refused, because many others will pay.
Competitive equilibrium, e.g. tacos at the flea market	**Many sellers, price-takers**. A stand asking more than $4 sells nothing, because shoppers go elsewhere.	**Many buyers, price-takers**. A shopper offering less than $4 finds no seller.

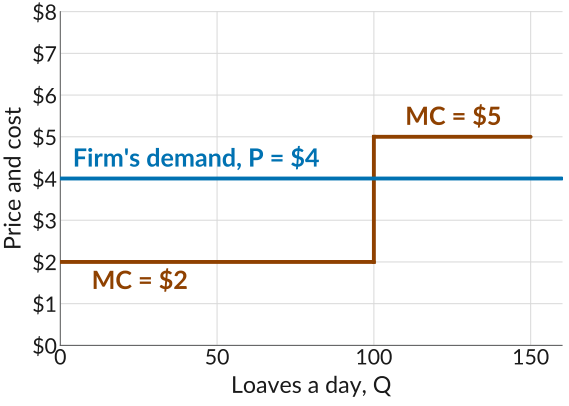
Firms That Take the Price as Given

Now look at one seller in a competitive market. How much should a price-taking firm produce?

- A firm selling a differentiated product faces a downward-sloping demand curve and chooses its price.
- When products are identical and buyers switch freely, a firm cannot charge more than the going price. It is a price-taker.
- **Example**: Sunrise Bakery is one of many bakeries in town selling the same sourdough loaf. The going price is $4 a loaf.
- Sunrise can bake up to 100 loaves a day at a marginal cost of $2, and up to 50 more on an overnight shift at $5 each. Its fixed costs are $120 a day.

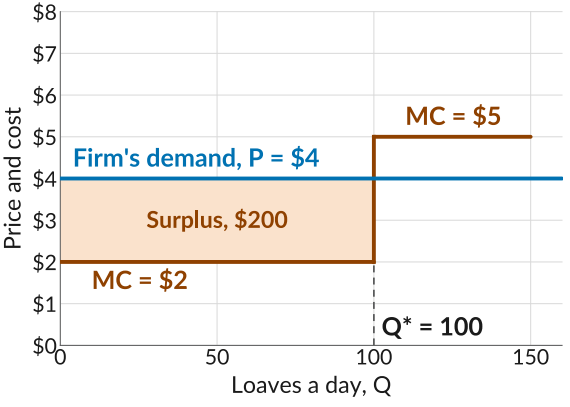
The Firm's Demand Curve

- Above $4, Sunrise sells nothing: customers walk to the next bakery. At $4, it can sell as many loaves as it likes.
- So its demand curve is flat at $4, and its only choice is how many loaves to bake.
- Its marginal cost steps up from $2 to $5 at 100 loaves.



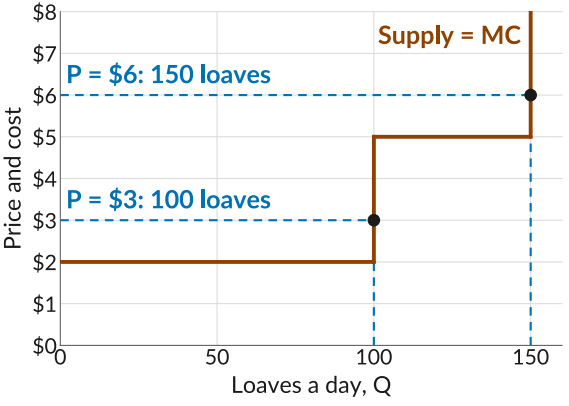
How Many Loaves?

- Each of the first 100 loaves adds $4 - 2 = 2$ to profit.
- Each overnight loaf would lose $5 - 4 = 1$.
- So Sunrise bakes $Q^* = 100$: every loaf whose marginal cost is below the price.
- Its surplus is $(4 - 2) \times 100 = 200$, the shaded area. Profit is $200 - 120 = 80$.



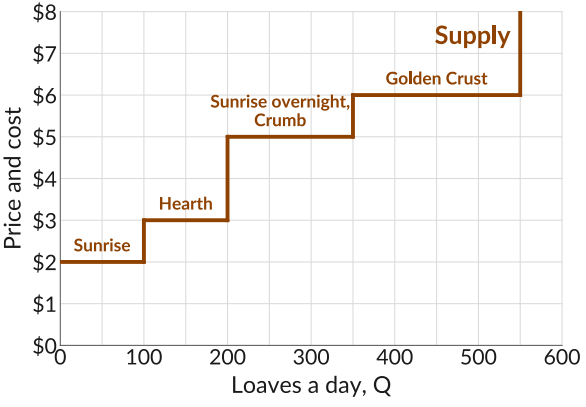
The Firm's Supply Curve

- A price-taking firm takes the market price and chooses a quantity by comparing it with marginal cost.
- Below $2, Sunrise bakes nothing. Between $2 and $5, it bakes 100. Above $5, it bakes 150.
- So the marginal cost curve is the firm's **supply curve**: at each price, the quantity it produces.



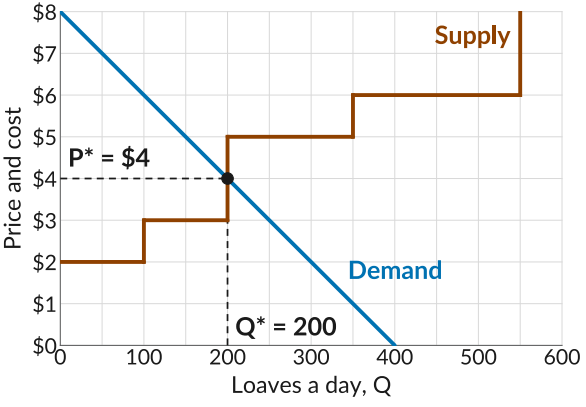
Adding Up the Bakeries

- Three more bakeries: Hearth bakes up to 100 loaves at $3, Crumb 100 at $5, and Golden Crust 200 at $6.
- **Market supply** adds up what every bakery bakes at each price, lowest marginal cost first. At $4, that is 200 loaves.
- Its height at each quantity is the marginal cost of that loaf: the 150th loaf costs $3.



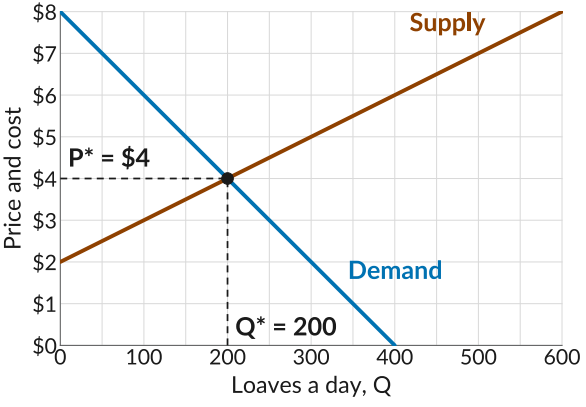
Competitive Equilibrium in the Bread Market

- Lined up highest first, the Qth buyer in town will pay $8 - Q/50$. At \$4, that is 200 buyers.
- Supply equals demand at $P^* = \$4$ and $Q^* = 200$. Sunrise and Hearth bake; Crumb and Golden Crust do not.
- The going price that Sunrise takes as given is set by supply and demand in the whole market.



With Many Bakeries

- With 50 bakeries instead of four, the supply curve has dozens of small steps, so we draw it as a smooth line.
- The market supply curve is the marginal cost curve for all the bread in town: its height at *Q* is the cost of the *Q*th loaf.
- In equilibrium, the 200th buyer's willingness to pay and the 200th loaf's marginal cost both equal the price, $4.



What to Do Next

- **Quiz 6 is Monday, October 26**, and covers Lectures 15 and 16.
- To prepare, go over the **slides**, **worksheets**, textbook readings (sections **8.1** to **8.4**), and **practice problems** for both lectures.